

Lab 3 – Architecture Selection & Justification

Architecture Selection

We chose **Layered Architecture for the Self-Service Coffee Kiosk System**.

Architectural Choice

The system is divided into separate layers: **Presentation, Business, and Data**. The User Interface handles touchscreen interactions, the Order Manager handles order processing, and the Database stores menu and pricing information.

Reason 1

Layered Architecture provides **clear separation of concerns**, making the different functions of the coffee kiosk easier to organize and maintain.

Reason 2

The kiosk has distinct responsibilities such as **touchscreen interaction, order management, payment processing, receipt printing, and data storage**, which can be organized into appropriate layers.

Security Advantage

Separating the **presentation, business, and data layers** helps prevent the user interface from directly accessing the stored data, providing better control over data access.

Performance Benefit

Separating responsibilities between layers allows the system to process **customer orders and data operations in an organized manner**, avoiding unnecessary interaction between unrelated components.